tissue, especially in liver, muscle and heart. In 1929, Von Gierke identified the first patient with GSD type I and named it as hepatonephromegalia glycogenica.The enzymatic defect, namely glucose 6 phosphatase deficiency, that caused type I GSD was elucidated in 1952 byCarl and Gerty Cori. They found that glucose 6-phosphatase was missing from the liver of a patient with this disease. This was the first demonstration of an inherited deficiency of a liver enzyme. The liver glycogen is normal in structure but present in abnormally large amounts.

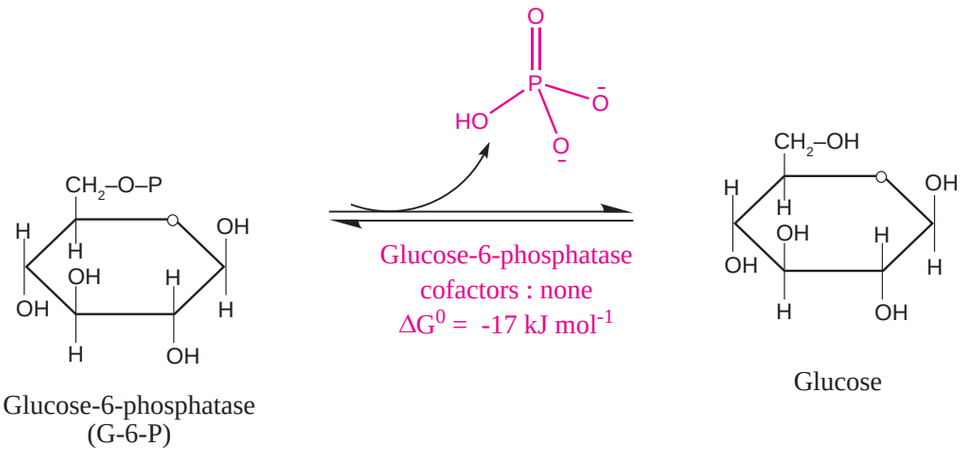

Figure.7.3 Action of glucose-6-phosphatase

Glucose -6-phophatase catalyzes the final step leading to the release of glucose in blood stream by the liver **(Figure.7.3)**. Deficiency of this enzyme results in an increase in intracellular glucose -6-phosphate. This phosphorylated sugar cannot leave the liver, because it cannot cross the plasma membrane, which leads to a large accumulation of glycogen in the liver and inability to increase blood glucose concentration in response to glucagon or epinephrine. Since glucose6- phosphate cannot leave liver cells, there is compensatory increase in glycolysis leading to increased levels of pyruvic acid and lactic acid.

Patients who have Von Gierke disease also have an increased dependence on fat metabolism. This disease can also be produced by a mutation in the gene that encodes the glucose 6-phosphate transporter.

7.2.1 Clinical manifestations

The clinical manifestations of Von Gierke disease, as shown in Figure.7.4, include hypoglycemia with lactic acidosis during fasting, enlarged liver, distended abdomen, cherubic or doll like face and seizures.

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